

Differential Effects are Absent During Acute Aerobic and Anaerobic Exercise on Working Memory in Older Adults

Anahita Goodarzi, Pascal Petherick, Alexandra Tenn, Erin Wills

Author Contributions

PP: Wrote/formatted results; researched for introduction; determined sampling procedure.

EW: Drafted sampling; generated visual/numerical summaries; contributed to statistical analyses of data; formatted and generated RMD file.

AT: Drafted introduction, discussion, references; contributed to writing of statistical analysis; searched and summarized relevant papers

AG : Drafted methods, materials, statistical analysis; wrote/refined results.

Introduction

This study examines the immediate effects of acute aerobic and anaerobic exercise on working and short-term memory in adults aged 66 and older. As individuals age, the brain undergoes structural and functional changes, including accelerated degeneration of white matter beginning in early aging (around age 66), which contributes to declines in cognitive function, particularly memory (Mousley et al., 2025; Deary et al., 2009). Memory impairments can negatively affect daily functioning and overall quality of life, with deficits observed across long-term, short-term, and working memory domains (Deary et al., 2009). While regular physical activity has been shown to mitigate long-term cognitive decline, cognition in older adults is not static and can fluctuate between periods of optimal functioning and temporary impairment, often described as “good days” and “bad days” (Rockwood et al., 2014). This raises an important question: can acute exercise improve memory performance during periods of transient cognitive difficulty?

Despite strong evidence supporting the long-term cognitive benefits of exercise, its immediate effects on memory, particularly in older adults, remain unclear. Studies in younger populations suggest that both aerobic and anaerobic exercise can improve working memory and attention, with aerobic exercise showing slightly greater benefits (Erwin & Schreiber, 2024). However, a systematic review by McSween et al. (2019) highlights the limited number and overall quality of studies examining acute exercise effects, emphasizing the need for more statistically rigorous research in this area. Understanding whether exercise can provide immediate cognitive benefits is important, as it may offer a practical and accessible intervention for managing short-term fluctuations in cognitive function among aging populations.

This study aimed to investigate the comparative effects of acute aerobic and anaerobic exercise on working and short-term memory in adults aged 66 and older. Based on prior evidence demonstrating cognitive benefits of exercise, particularly in younger populations where aerobic exercise has shown slightly greater improvements in working memory, we hypothesized that both exercise groups would outperform the control group, with the aerobic group demonstrating the greatest improvement in mean performance. Contrary to this hypothesis, no statistically significant differences were observed between treatment groups across all memory outcomes.

Materials and Methods

Sampling and Study Design

To test this hypothesis, we conducted an experiment comparing the effects of aerobic and anaerobic exercise on memory performance across three treatment groups: aerobic, anaerobic, and control.

Participants were selected using a stratified random sampling approach. Three villages were randomly chosen from each island, and birth records from years 300–311 were used to randomly select 11 participants per village, yielding a total sample of 99 participants aged 66 years and older. Individuals who were de-

ceased or declined participation were replaced through repeated random selection. This approach improved representativeness, although some selection bias may remain.

A randomized controlled trial design was implemented. Participants were randomly assigned to one of three groups: aerobic exercise ($n = 33$), anaerobic exercise ($n = 33$), or no-exercise control ($n = 33$). The aerobic group completed two 30-minute indoor walking sessions, while the anaerobic group completed three 20-minute high-intensity interval training (HIIT) sessions. Control participants did not perform any exercise.

Prior to testing, participants completed a brief survey assessing potential confounding variables, including sleep duration (hours), fatigue (rated on a scale from 1 to 10), and recent physical activity (minutes per week). Participants were not stratified by gender, as previous research suggests that sex differences in cognitive decline and memory performance in older adults are small and not consistently statistically significant (Yaffe et al., 2009; McDougall et al., 2014).

Data Collection & Measurements

Participants completed three memory tasks immediately following exercise (or following the survey for the control group). Short-term memory was assessed using a card recall test (10 cards) and a vocabulary recall test (20 words). Working memory was assessed using a card-matching memory game involving 30 face-down cards, with performance measured as time (seconds) to completion.

To assess potential confounding variables, participants were asked the following survey questions prior to testing:

- “How many hours did you sleep last night?”
- “On a scale of 1 to 10, how tired do you feel right now?”
- “How many minutes did you spend doing low-, moderate-, or vigorous-intensity physical activity last week?”

These variables were recorded to assess potential confounding but were not explicitly controlled for in the primary analysis due to limitations of the Islands simulation.

During the study, two participants in the control group passed away after group assignment but prior to data collection. As a result, the final sample size was $n = 97$, with slightly unequal group sizes (control $n = 31$; aerobic and anaerobic $n = 33$). The implications of this imbalance are addressed in the Discussion.

Statistical Analysis

Data were analyzed using inferential statistics at a significance level of $\alpha = 0.05$. A one-way ANOVA was used to assess differences in mean performance across treatment groups for each memory outcome. When appropriate, Tukey’s Honest Significant Difference (HSD) test was conducted for pairwise comparisons between groups.

The assumptions of ANOVA were assessed using graphical summaries. Distributions within groups appeared approximately symmetric with no extreme outliers, and variability across groups was similar, supporting the use of ANOVA.

To complement these analyses, 95% bootstrap confidence intervals were constructed to estimate uncertainty in mean performance across treatment groups. Bootstrap sampling with replacement was performed 1000 times for each treatment-outcome combination, using sample sizes corresponding to each group ($n = 31$ for control; $n = 33$ for aerobic and anaerobic). The 2.5th and 97.5th percentiles of the bootstrap distributions were used to define confidence intervals.

Results

Across all memory outcomes, no statistically significant differences were observed between treatment groups (all p -values > 0.05). Measures of central tendency (means/medians) and variability were similar across

aerobic, anaerobic, and control groups, indicating minimal differences in overall performance.

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## data$treatment.group  2    1150    575.2    2.187  0.118
## Residuals           94   24719    263.0
```

Figure 1. One-way ANOVA Test Comparing Memory Game Performance Across Treatment Groups

Figure 1 depicts a one-way ANOVA test conducted on the memory game completion time of the individuals in each treatment group. The analysis showed no statistically significant effect of exercise type on performance ($F = 2.187$), indicating insufficient evidence to conclude that mean performance differed between groups. The between-group variation was small relative to within-group variability, contributing to the non-significant ANOVA result. Using $\alpha = 0.05$, there is insufficient evidence to reject the null hypothesis that any of the mean memory game performances differ significantly ($p = 0.118 > 0.05$.)

The assumptions of ANOVA were assessed and met. Memory game performance data were continuous and independent, with approximately normal distributions within each group (Q-Q plots used to assess normality can be found in the appendix). Additionally, it was found that none of the SDs differ significantly ($15.96/15.11 < 2$, $17.48/15.11 < 2$). Thus, ANOVA was determined to be an appropriate procedure.

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = data$memory.game.seconds ~ data$treatment.group)
##
## $`data$treatment.group`
##      diff      lwr      upr      p adj
## Anaerobic-Aerobic  7.6090909 -1.897949 17.116131 0.1426194
## Control-Aerobic    6.8453568 -2.813806 16.504519 0.2152515
## Control-Anaerobic -0.7637341 -10.422896  8.895428 0.9806482
```

Figure 2. Tukey's Honest Significant Difference Test for Pairwise Comparisons of Memory Game Performance

Figure 2 presents the results of Tukey's HSD post hoc analysis. No statistically significant pairwise differences were observed between any treatment groups (adjusted p-values > 0.05). Notably, there was no significant difference between the aerobic and anaerobic groups ($p = 0.14$), suggesting that exercise type did not differentially affect memory game performance.

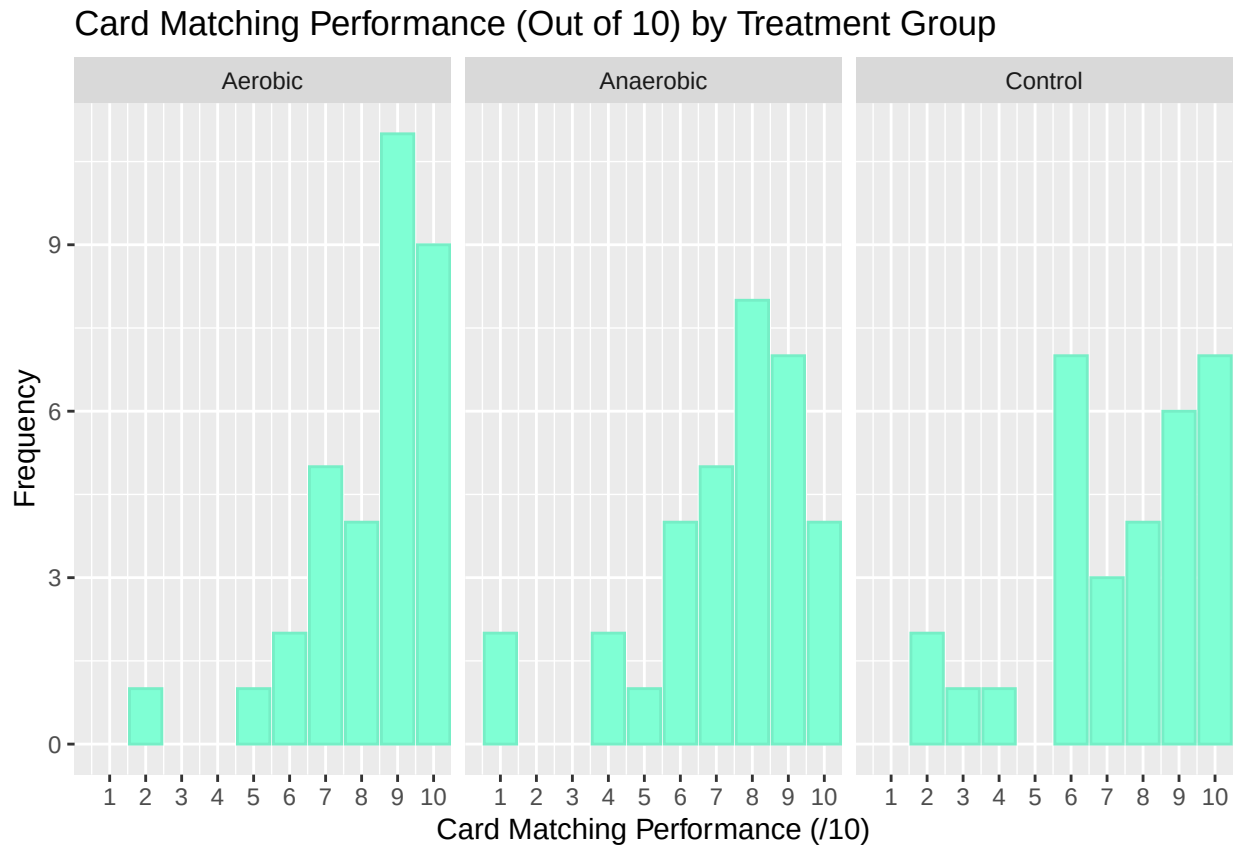


Figure 3. Card Matching Performance (/10) by Treatment Group

Figure 3 shows the distribution of Card Matching test scores for participants in the aerobic, anaerobic, and control groups. The bar graphs display the frequency of scores across the possible range of values for each treatment group. Performance was similar across all groups, with substantial overlap in score distributions. While the aerobic group showed slightly higher frequencies of higher scores, these differences were small and not clearly distinct from those observed in the anaerobic and control groups.

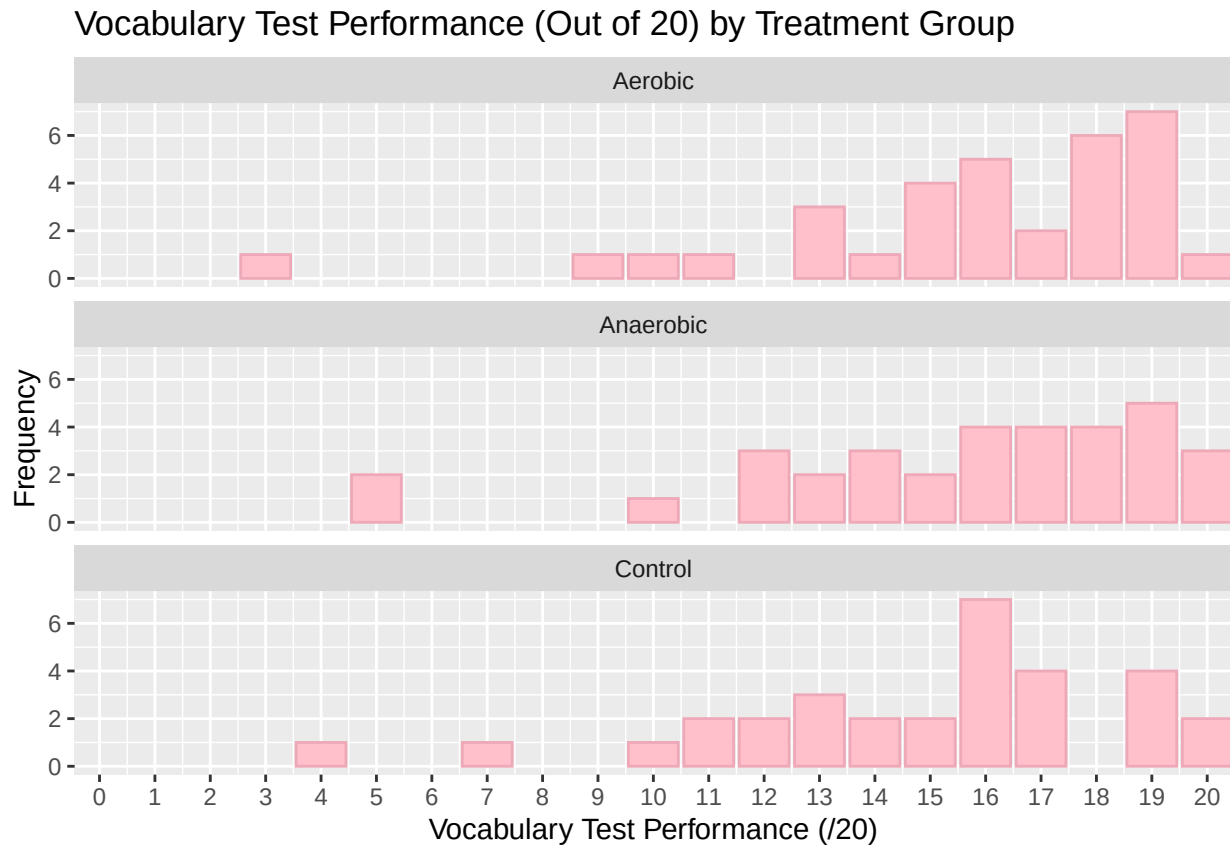


Figure 4. Vocabulary Test Performance (/20) by Treatment Group

Figure 4 presents the distribution of Vocabulary test scores for the aerobic, anaerobic, and control groups. Each bar graph illustrates the frequency of participant scores within each treatment group. As with the card recall task, score distributions were highly similar across groups, with overlapping ranges and no clear separation in performance. Any observed differences in frequencies were and did not suggest meaningful differences between treatment conditions.

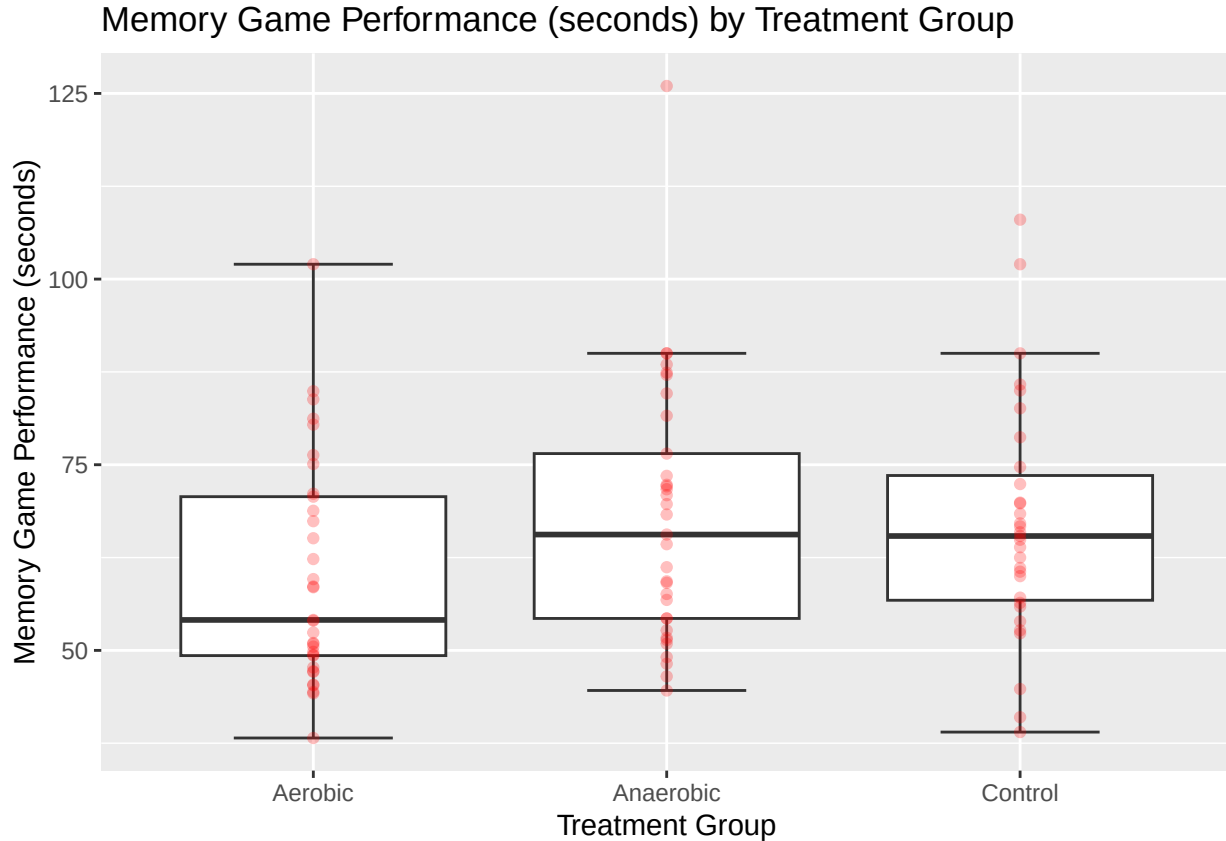


Figure 5. Memory Game Performance (s) by Treatment Group

Figure 5 shows boxplots representing the distribution of Memory Game completion times (in seconds) for participants in the aerobic, anaerobic, and control groups. The boxplots illustrate the median, interquartile range, and potential outliers for each treatment group. Points indicating the values of each datum are additionally present. Median performance and interquartile ranges were comparable across groups, with no clear differences in central tendency or variability. The aerobic group showed slightly lower completion times on average, most clearly illustrated in the card recall test, which scored short-term memory (see figure 3). Anaerobic treatment group participants also averaged slightly higher than their control participant counterparts, although this difference was even more subtle. These differences were small relative to the spread of the data and consistent with the non-significant ANOVA results. Table 2 contains no p-values less than 0.05.

	Card recall (/10)	Vocab recall (/20)	Memory game (s)
Aerobic	7.7 - 8.9	14.5 - 16.9	55.4 - 65.7
Anaerobic	6.48 - 8.00	14.2 - 16.8	62.2 - 74.2
Control	6.5 - 8.2	13.5 - 16.0	61.9 - 73.1

Table 1: 95% Bootstrap Confidence Intervals for the Average Performance on Memory Tests by Treatment Group.

Bootstrap samples were performed on the basis that original samples were representative of the true Islands population of individuals aged 66 or older, and also that $n > 30$ for each treatment group. The substantial overlap in 95% bootstrap confidence intervals across all treatment groups suggests no meaningful differences in mean performance. The lack of differences in means undermines the influence of exercise, particularly aerobic exercise, on memory performance. While the aerobic group showed slightly higher upper bounds in the Card and Vocabulary Recall tests (indicating higher scores), and slightly lower bounds in the Memory

Game test (indicating faster completion time) these differences are small and fall within the range of variability seen in the other groups. This overlap indicates that any observed differences in sample means are small relative to within-group variability and are not consistent with a meaningful effect of exercise type on memory performance.

	Card recall (/10)	Vocab recall (/20)	Memory game (s)
$\mu_{ctrl} - \mu_{anaerobic}$	p = 0.858	p = 0.445	p = 0.854
$\mu_{ctrl} - \mu_{aerobic}$	p = 0.099	p = 0.308	p = 0.097
$\mu_{aerobic} - \mu_{anaerobic}$	p = 0.041	p = 0.793	p = 0.053

Table 2: P-Values From Randomization Test (n=1,000) Representing Likelihood of Obtained Experimental Outcomes Assuming the Null Hypothesis is True.

The randomization tests in Table 2 assume that participants were randomly assigned to the aerobic, anaerobic, and control groups, ensuring independence between observations and reducing systematic bias. This condition is met because there was an equal number of participants in each treatment group. Randomization helps ensure that potential confounding variables, such as prior fatigue or exercise history, are approximately balanced across groups. Additionally, randomization tests do not require normally distributed data, making them appropriate for the card and vocabulary recall tests. Therefore, the assumptions required for randomization-based inference are satisfied.

The p-values for comparisons between the aerobic and control groups and between the anaerobic and control groups all exceeded the significance threshold of $\alpha = 0.05$, indicating insufficient evidence to reject the null hypothesis for these comparisons. Although the randomization test indicated a statistically significant difference between the anaerobic and aerobic groups for the card recall task (p = 0.041), this result was not consistent across other memory outcomes and was not supported by the ANOVA or Tukey analyses. Therefore, this isolated finding is likely attributable to sampling variability rather than a robust effect of exercise type on memory performance.

While the results obtained did follow a pattern that agrees with the hypothesis, the aerobic treatment group outperformed the anaerobic, and the anaerobic outperformed the control, these differences in performance were determined to be statistically negligible. Inferential analyses did not support statistically significant differences between treatment groups.

Conclusions

Acute aerobic and anaerobic exercise did not produce statistically significant improvements in working or short-term memory in adults aged 66 and older. Although descriptive trends suggested slightly better performance in the aerobic group, inferential analyses, including ANOVA and bootstrap confidence intervals, indicated that these differences were not statistically significant. These findings suggest that acute exercise may not be an effective immediate intervention for improving memory performance in older adults.

Although descriptive trends indicated slightly better performance in the aerobic group, these findings were not supported by the inferential analyses. These results suggest that any immediate cognitive benefits of exercise in this population are likely small or highly variable. Further research with larger sample sizes is needed to determine whether acute exercise has a meaningful clinical impact on memory performance in older adults.

Discussion

This study found no statistically significant differences in working or short-term memory performance between aerobic, anaerobic, and control groups, contrary to our hypothesis. Although descriptive trends suggested slightly better performance in the aerobic group, these differences were small and likely attributable to sampling variability rather than a true effect. Post hoc comparisons using Tukey’s Honest Significant Difference (HSD) test indicated no statistically significant differences between any treatment groups, including

between the aerobic and anaerobic groups, suggesting that exercise type did not differentially influence memory performance. One possible explanation is that acute exercise produces only modest or highly variable cognitive effects in older adults, which may not be detectable in a single-session experimental design.

These findings contrast with the well-established long-term cognitive benefits of exercise reported in the literature. While regular physical activity is associated with structural and functional brain adaptations that support cognitive function, acute bouts of exercise may not produce immediate measurable improvements in memory performance. Previous studies have reported modest short-term cognitive benefits of exercise, particularly in younger populations and in working memory tasks (Chang et al., 2012; Erickson et al., 2011). Similarly, aerobic exercise has been shown to produce slightly greater cognitive benefits than anaerobic exercise (Erwin & Schreiber, 2024), which is consistent with the descriptive trends observed in this study. However, unlike these studies, our results did not reach statistical significance. This discrepancy may be explained by differences in study populations and design, as acute exercise effects in older adults are less consistent and often exhibit greater variability (Colcombe & Kramer, 2003; McSween et al., 2019).

Several limitations should be considered when interpreting these results. First, the sample size was relatively small at the group level, which may have limited statistical power to detect subtle differences between treatment groups. Second, group sizes were slightly unequal due to participant loss in the control group, which may have introduced minor imbalance. Third, potential confounding variables such as sleep, fatigue, and prior physical activity were measured but not controlled for in the analysis, which may have introduced additional variability in performance. Finally, the simulated nature of the Islands environment may limit the generalizability of these findings to real-world populations.

Although no statistically significant effects were observed, the results may still have practical relevance. Even small improvements in memory performance could be meaningful for older adults experiencing cognitive decline. This highlights the distinction between statistical significance and clinical significance, where effects that are not statistically detectable may still be practically important.

Overall, these findings suggest that acute exercise is unlikely to be a reliable short-term intervention for improving memory performance in older adults. Future research should aim to increase sample size, improve control of confounding variables, and explore repeated or longer-duration exercise interventions. Incorporating baseline cognitive assessments or stratifying participants by fitness level may also help identify subgroups that may benefit more from acute exercise.

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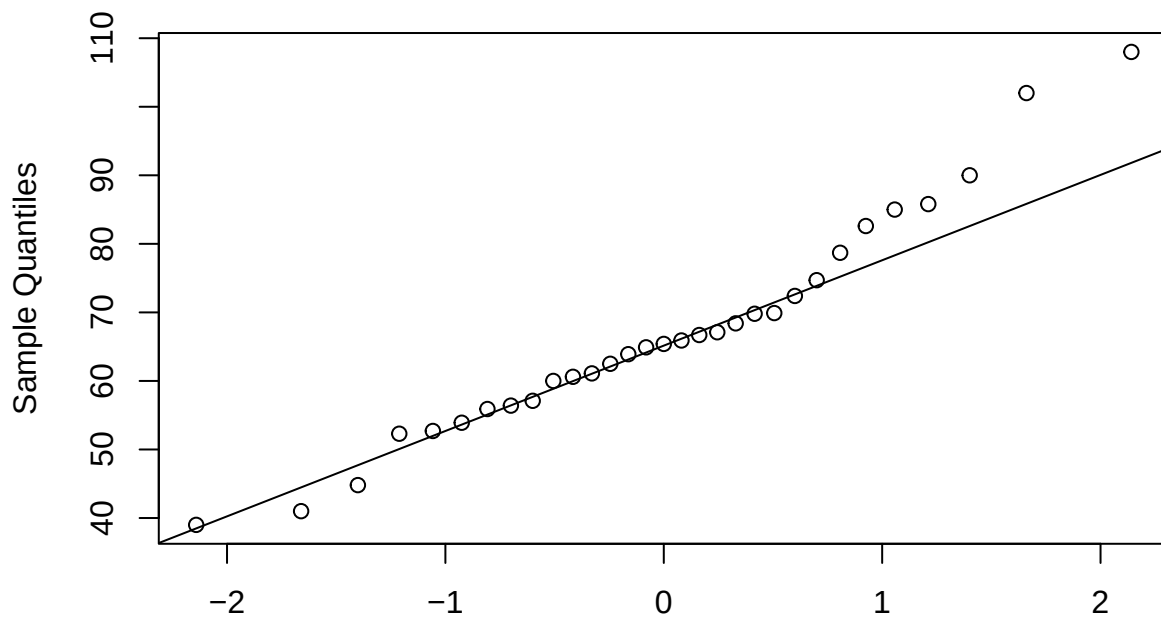
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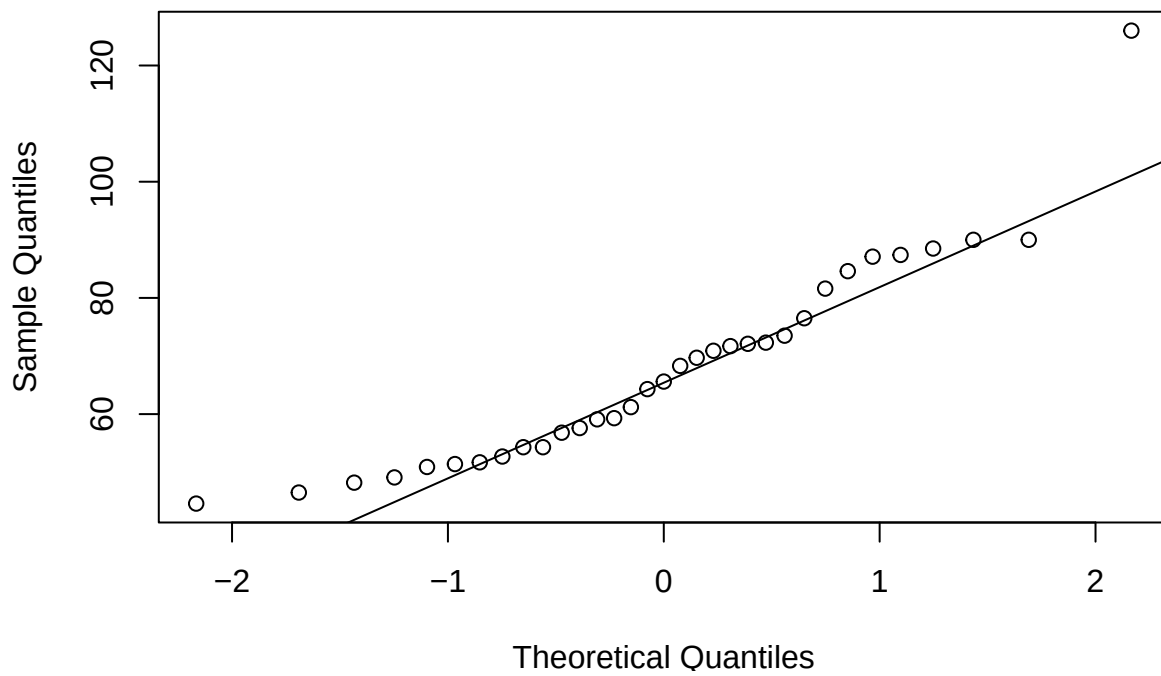
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Appendixes

Normal Q-Q Plot



Normal Q-Q Plot



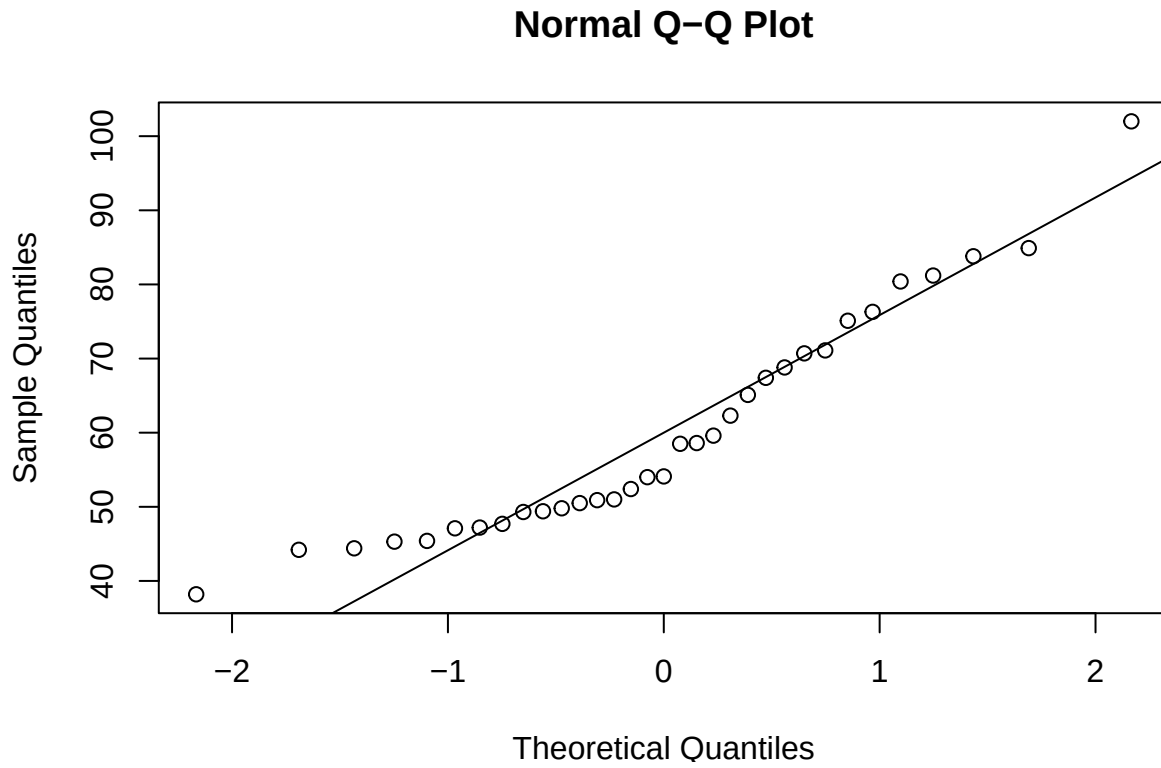


Figure 6, 7, 8. Q-Q Plots to Assess Normality of Memory Test Data.



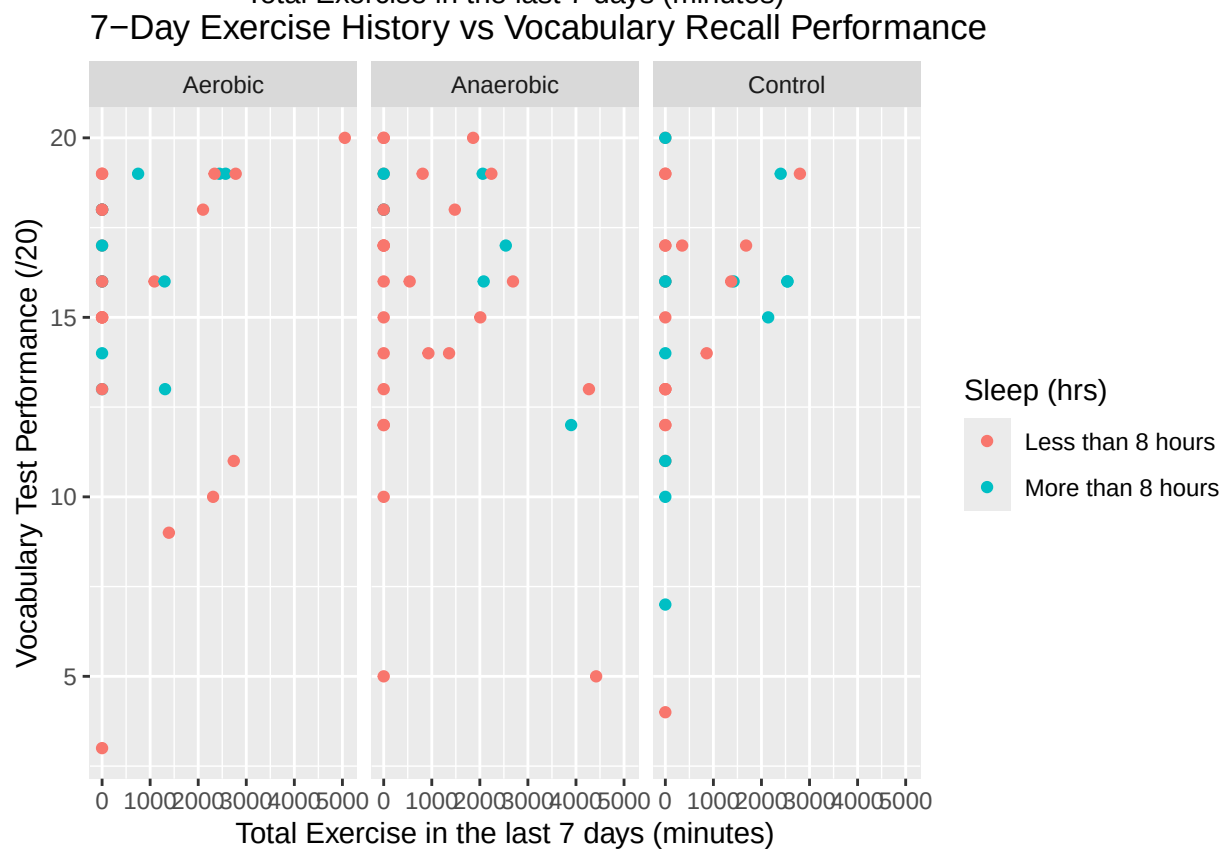


Figure 9, 10, 11. Scatterplots comparing 7-day exercise history and test performance (mem-

ory/card/vocabulary). Separated by treatment group. Filled by hours of sleep (</> 8 hours.)

	Memory	Cards	Vocabulary
Control	67.05	7.42	14.84
Aerobic	60.20	8.33	15.79
Anaerobic	67.81	7.30	15.55

Table 3. Table comparing the means for each treatment-test pairing.

	Memory	Cards	Vocabulary
Control	15.96	2.36	3.70
Aerobic	15.11	1.78	3.61
Anaerobic	17.48	2.28	3.81

Table 4. Table comparing the standard deviations for each treatment-test pairing.

	Memory	Cards	Vocabulary
Control	65.4	8	16
Aerobic	54.1	9	16
Anaerobic	65.6	8	16

Table 5. Table comparing the medians for each treatment-test pairing.

	Memory	Cards	Vocabulary
Control	39.0-108.0	2-10	4-20
Aerobic	38.2-102.0	2-10	3-20
Anaerobic	44.6-126.0	1-10	5-20

Table 6. Table comparing the ranges for each treatment-test pairing.

Q1:

	Memory	Cards	Vocabulary
Control	56.75	6	13
Aerobic	49.3	7	15
Anaerobic	54.3	6	14

Table 7. Table comparing the 1st quartile for each treatment-test pairing.

Q3:

	Memory	Cards	Vocabulary
Control	73.55	9	17
Aerobic	70.7	10	18
Anaerobic	76.5	9	18

Table 8. Table comparing the 3rd quartile for each treatment-test pairing.